# Reading Between the Lines: LLM-Based Label Imputation and Meta-Learning for Noisy Labels in Emergency Department Triage

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**Abstract.** Children presenting to the SickKids pediatric emergency department wait twice: once for physician assessment, and again for diagnostic test results. The Machine Learning Medical Directive (MLMD) system eliminates the second wait for high-confidence cases by predicting required tests at triage and authorizing them in parallel. Each missed prediction returns the patient to the slower, serialized workflow, making recall the primary performance target. Label noise constrains this recall. Tests ordered at external facilities go unrecorded in the SickKids electronic health record (EHR), causing genuine test-ordering events to appear as negatives and teaching the model spurious clinical absences.

We address this across 483,705 pediatric ED encounters spanning June 2018 to January 2026. At the data level, we replace the production regex imputation system with a clinically validated Knowledge Graph and GraphRAG retrieval pipeline that recovers 11,110 previously unrecorded test events across four diagnostic pathways. At the training level, we apply LiLAW (Lightweight Learnable Adaptive Weighting) [1], a meta-learning framework that down-weights residual false negatives through bilevel optimization over three learned scalar parameters.

GraphRAG imputation alone increased recall by **[TODO]%**. The combined system achieved recall of **[TODO]%** and precision of **[TODO]%**, recovering **[TODO]%** of missed test authorizations while maintaining deployment-required precision thresholds. Knowledge Graph imputation paired with noise-aware meta-learning offers a deployable path to correcting asymmetric EHR label noise without cross-institutional data integration.

**Keywords:** *label noise, clinical decision support, machine learning, label imputation, meta-learning*

## Introduction and Clinical Motivation

At The Hospital for Sick Children (SickKids), children presenting to the emergency department (ED) wait twice: once for a physician to assess them, and again for test results to return. The Machine Learning Medical Directive (MLMD) system [2] eliminates the second wait by predicting which diagnostic tests a patient will need at triage and authorizing them before the physician arrives. A test authorized at triage completes while the patient is still waiting to be seen; a test ordered only after assessment adds another full cycle. The system targets four high-volume pathways: arm X-rays, electrocardiograms (ECGs), appendix ultrasounds, and testicular ultrasounds.

A structural gap in the SickKids electronic health record (EHR) limits what the model can learn. When a patient is transferred to, or later evaluated at, an external facility, any test ordered there never appears in SickKids' records. The hospital's EHR is not integrated with external provider systems. The model observes these missing tests as negatives, and because the missingness tracks transfer patterns rather than clinical need, it learns systematic errors: associating certain presentations with "test not needed" when the test was simply performed elsewhere. The production system attempts to correct this with regular-expression matching over clinical notes, but regex cannot capture semantic variants or clinical shorthand, leaving a large class of false negatives in the training data. Across 483,705 triage encounters spanning June 2018 to January 2026, each of these silent absences trains the model toward the same wrong generalization: this presentation does not require testing.

We address this noise at two levels. At the data level, we replace regex matching with a dynamically constructed Knowledge Graph and GraphRAG retrieval pipeline that resolves ambiguous negatives against clinical guidelines before training. At the training level, we apply LiLAW (Lightweight Learnable Adaptive Weighting) [1], a meta-learning framework that down-weights suspected false negatives the imputation step cannot adjudicate. Together, these target asymmetric noise without requiring cross-institutional EHR integration or access to ground-truth transfer records. The Knowledge Graph pipeline recovered 11,110 test-ordering events the regex baseline missed; the experiments below assess whether those additional labels, combined with noise-aware training, translate to recall gains at deployment.

## Related Work

*Label imputation in EHRs.* Missing labels in clinical datasets are rarely missing at random; their absence reflects institutional silos [3] and clinical workflow patterns [4]. Most imputation work targets missing continuous *features* (vital signs, laboratory values) [5], but imputing missing *labels* is a distinct problem. The SickKids noise structure fits the Positive-Unlabeled (PU) learning paradigm [6]: positive labels are trustworthy, while the unlabeled class conflates true negatives with unobserved external events. Prior PU approaches treat the unlabeled class as a statistical quantity to estimate, whether through Bayesian mixture models [7] or label-recovery frameworks for clinical codes [8]. Our approach resolves the ambiguity directly: we query a validated Knowledge Graph against each patient's post-assessment diagnosis before training begins, converting a probabilistic inference problem into a clinical evidence lookup where the graph permits it.

*Knowledge graphs and LLMs in clinical decision support.* LLMs show strong capacity for medical knowledge imputation [9] and clinical entity recognition from unstructured notes [10], but standalone LLM outputs lack auditability and hallucinate in high-stakes settings. Retrieval-augmented generation (RAG) over clinical guidelines improves extraction accuracy by grounding outputs in validated sources [11, 12]. GraphRAG architectures [13, 14] extend this by combining structured Knowledge Graph reasoning with contextual retrieval, enabling semantically coherent inference across interconnected clinical concepts. We adopt a GraphRAG architecture but depart from existing work in three ways: the Knowledge Graph is constructed dynamically from live clinical guidelines, enriched with ontology-standardized codes and quantitative literature evidence, and every diagnosis-to-test pathway is validated through a multi-agent clinical audit before use.

*Meta-learning for noisy labels.* Early approaches to learning with noisy labels modified the loss function itself. Loss-correction methods estimate a class-conditional noise transition matrix and use it to adjust either the loss or the model's predictions, yielding an unbiased estimator of the clean risk under ideal conditions [15]. Generalized Cross Entropy [16] and Symmetric Cross Entropy [17] avoid explicit matrix estimation by interpolating between

noise-tolerant and optimization-friendly loss families. These methods improve robustness when corruption follows class-level flip rates, but they impose a fixed rule at the loss level and cannot adapt per-sample trust as training progresses.

Sample selection methods addressed this limitation by choosing which examples to learn from at each step. MentorNet [18] learns a data-driven curriculum through an auxiliary network, typically requiring clean supervision to train the mentor. Co-teaching [19] removes the clean-data assumption by training two peer networks that exchange small-loss examples, at roughly twice the backbone cost. DivideMix [20] extends this idea by fitting a two-component Gaussian mixture to per-sample losses and recasting suspected-noisy examples as unlabeled data for semi-supervised co-training. These methods exploit the empirical finding that deep networks fit clean patterns before memorizing noise, but low loss is an imperfect proxy for label correctness. Under asymmetric corruption, genuinely difficult examples can resemble mislabeled ones, and the heuristic breaks down.

Meta-learning reweighting made the selection rule itself learnable through bilevel optimization. Learning to Reweight Examples (L2RW) [21] optimizes one scalar weight per training example in each mini-batch to minimize loss on a small clean validation set, at roughly three times standard training cost. Meta-Weight-Net [22] replaces per-example weights with a small multilayer perceptron that maps loss to weight, but retains the clean-validation-set requirement. Confident Learning [23] takes a complementary approach, estimating the joint distribution of noisy and true labels to prune errors before training rather than reweight during it. LiLAW [1] compresses the meta-learner to three global scalar parameters ($\alpha$, $\beta$, $\delta$) that weight easy, moderate, and hard samples based on softmax agreement and prediction confidence. It retains bilevel updates against validation mini-batches but does not require those labels to be clean or unbiased, and adds no auxiliary networks. These properties match the SickKids setting, where a small trustworthy validation set may not exist and the noise structure (unrecorded external tests producing false negatives, not symmetric random flips) differs from the corruption patterns most benchmark methods were designed for.

## Data and Setting

The dataset comprises EHR records from pediatric patients (aged 0–18 years) presenting to the SickKids ED between June 2018 and January 2026. After excluding patients who left without being seen ($n$ = 25,397) and encounters missing the free-text clinical diagnoses required for label imputation ($n$ = 11,396), the final cohort contained $N$ = 483,705 unique ED encounters (median age 5.0 years, IQR 2.0–10.0; 54.1% male).

The model's feature space is limited to information available at triage: demographics (age, sex, weight), vital signs (pulse, respiration, temperature, blood pressure), chief complaint narrative, and Canadian Triage and Acuity Scale (CTAS) score. Post-assessment ICD-10 diagnosis codes, assigned by the treating physician after evaluation, are used only by the label imputation pipeline and excluded from the model's input features to prevent target leakage.

Four binary test-ordered labels define the prediction targets: Arm X-Rays ($n$ = 1,904 positive encounters), ECGs ($n$ = 12,236), Appendix Ultrasounds ($n$ = 3,293), and Testicular Ultrasounds ($n$ = 2,889). Positive labels are reliable ground truth: they record tests ordered and performed at SickKids. Negative labels are ambiguous. A recorded absence may reflect a genuine clinical decision not to test, or a test performed at an external facility whose records never reach the SickKids EHR. This asymmetric noise structure is the central problem the methods below address.

## Methods

### Label imputation via Knowledge Graph

The label imputation pipeline constructs an evidence-grounded Knowledge Graph (KG) mapping ICD-10 diagnosis codes to clinically indicated diagnostic tests. The KG serves as audited clinical ground truth: when a patient's post-assessment diagnosis maps to a test but the encounter lacks a positive label for that test, the label is imputed positive. Imputation runs offline over the training set only. No diagnosis codes are available at inference time.

*Knowledge Graph construction.* The graph is built in four stages.

*Stage 1: Baseline rule generation.* GPT-5-mini generates triage rules mapping clinical observations to conditions and their required tests. To reduce hallucination, the model is grounded via RAG over the EPFL clinical guidelines dataset, with retrieved guideline texts providing in-context evidence.

*Stage 2: Ontology standardization.* All entities are mapped to recognized medical ontologies through external APIs. Symptom nodes receive Human Phenotype Ontology (HP) and MONDO terms (EMBL-EBI OLS4 API); condition nodes are annotated with WHO ICD-10 codes (UMLS REST API) and ICD-10-CA codes (Canada Health Infoway FHIR); diagnostic test nodes receive LOINC panel codes.

*Stage 3: Evidence enrichment.* Edges are weighted with quantitative literature evidence. Peer-reviewed article and patent co-occurrence counts (Europe PMC API) attach to symptom-to-condition and condition-to-test edges; clinical trial counts (ClinicalTrials.gov) attach to treatment pathways. This allows the system to distinguish well-supported clinical associations from marginal correlations.

*Stage 4: Ontological expansion.* The graph is extended by integrating ClinGraph [24], a 153K-node, ~3M-edge clinical knowledge graph spanning SNOMED CT, ICD-10-CM, LOINC, RxNorm, and related vocabularies. ClinGraph back-fills missing cross-vocabulary codes and adds up to 500 first-degree neighbor nodes per existing node.

*Multi-agent audit.* Before imputation, every diagnosis-to-test pathway is validated by a four-persona multi-agent audit pipeline. The unit of review is a clinical triad: a (finding → condition → test) path extracted by traversing `INDICATES_CONDITION` followed by `REQUIRES_TEST` edges. Triads are deduplicated and sorted by clinical acuity (HIGH, MODERATE, STANDARD) to prioritize time-sensitive pathways.

Three independent personas evaluate each triad: a Protocol & Triage Enforcer with RAG access over clinical guidelines, a Rapid Diagnostics Researcher querying live literature databases (Semantic Scholar, Europe PMC), and an ED Attending evaluating clinical flow pragmatics and never-event mitigation. A fourth persona, the ED Medical Director, synthesizes these assessments into a verdict (*Verified*, *Flagged for Review*, or *Rejected*) and a continuous confidence score in $[0, 1]$. Three hard-flag conditions bypass scoring and unconditionally produce a *Flagged* verdict: a genuine clinical error in the pathway, a Choosing Wisely flag, or zero guideline and literature support. Audit outcomes are written back as edge attributes and inform retrieval during imputation.

*Imputation strategies.* We evaluate two retrieval strategies. *KG String Matching* uses bidirectional substring matching to map patient diagnosis strings to KG nodes, then traverses 1-hop (`REQUIRES_TEST` or `DIRECTLY_INDICATES_TEST`) and 2-hop (`INDICATES_CONDITION` → `REQUIRES_TEST`) edges to identify indicated tests. A minimum-node threshold $\tau$ requires at least $\tau$ independently matched nodes to nominate a test before imputation, reducing false positives from broad symptom matches. *GraphRAG Semantic Retrieval* embeds KG nodes and patient diagnoses using the MedEmbed biomedical language model, querying a local ChromaDB vector store to capture cases without lexical overlap. A sliding-window negation filter discards matches where the clinical concept is negated in the source text (e.g., "no chest pain" does not trigger ECG imputation). Indicated tests are ranked by aggregate cosine similarity across matched nodes and filtered by a threshold $\theta$.

### Noise-aware training via LiLAW

Label imputation corrects false negatives where a post-assessment diagnosis conclusively implies a test was needed. It cannot resolve cases where the clinical indication is ambiguous: chest pain sometimes warrants an ECG and sometimes does not, so no imputation rule can adjudicate these encounters. Residual false negatives of this kind remain in both the original and imputed training sets. LiLAW (Lightweight Learnable Adaptive Weighting) [1] handles this residual noise at training time by learning to down-weight samples whose observed label is inconsistent with the model's predictions.

*Sample difficulty characterization.* Let $s_i = \text{softmax}(f_\theta(x_i))$ be the model's output distribution for training sample $i$ with observed label $\tilde{y}_i$. LiLAW characterizes each sample by two quantities: $s_i[\tilde{y}_i]$, the predicted probability assigned to the observed label, and $\max(s_i)$, the model's peak class confidence. Their relationship defines four difficulty regimes:

- *Easy*: $s_i[\tilde{y}_i] = \max(s_i)$ and $\max(s_i)$ is high. The model agrees with the observed label confidently.
- *Moderate*: $\max(s_i)$ is low regardless of agreement. The model is uncertain.

- *Hard*: $s_i[\tilde{y}_i] < \max(s_i)$ and $\max(s_i)$ is high. The model confidently predicts a different class than the observed label — the signature of a likely mislabelled sample.

In our setting, hard samples correspond precisely to the false negatives that remain after imputation: encounters where the model assigns high probability to "test needed" but the training label records "test not performed." These are the cases most likely to reflect external-provider tests that were never captured in the SickKids EHR.

*Weight functions.* LiLAW introduces three scalar meta-parameters, $\alpha$, $\beta$, and $\delta$, each governing a weight function over the $(s_i[\tilde{y}_i], \max(s_i))$ plane:

$$\mathcal{W}_\alpha = \sigma(\alpha \cdot s_i[\tilde{y}_i] - \max(s_i)) \tag{1}$$

$$\mathcal{W}_\beta = \sigma(-(\beta \cdot s_i[\tilde{y}_i] - \max(s_i))) \tag{2}$$

$$\mathcal{W}_\delta = \exp\left(-\frac{(\delta \cdot s_i[\tilde{y}_i] - \max(s_i))^2}{2}\right) \tag{3}$$

$\mathcal{W}_\alpha$ is a sigmoid that is high for easy samples (agreement, high confidence); $\mathcal{W}_\beta$ is an inverted sigmoid that is high for hard samples (disagreement, high confidence); $\mathcal{W}_\delta$ is a radial basis function centered on the moderate regime (low confidence either way). The total sample weight is $\mathcal{W} = \mathcal{W}_\alpha + \mathcal{W}_\beta + \mathcal{W}_\delta$, and the LiLAW loss is $\mathcal{L}_W = \mathcal{W} \cdot \mathcal{L}$. As $\alpha$, $\beta$, and $\delta$ are learned, the model adapts which difficulty regime to emphasise at each stage of training rather than applying a fixed curriculum.

*Bilevel optimization.* The meta-parameters are learned via bilevel optimization. Each training mini-batch alternates between two steps. In the inner step, $\alpha$, $\beta$, and $\delta$ are frozen and the model parameters $\theta$ are updated by gradient descent on the weighted training loss $\mathcal{L}_W$. In the outer step, $\theta$ is frozen and a validation mini-batch is drawn; $\alpha$, $\beta$, and $\delta$ are then updated by SGD on the LiLAW loss evaluated on that validation batch. The joint objective is:

$$\begin{aligned} \theta^* &= \arg\min_{\theta} \sum_{(x_i, \tilde{y}_i) \in \mathcal{D}_t} \mathcal{L}_{W_{\alpha^*, \beta^*, \delta^*}} \\ \text{s.t.} \quad \alpha^*, \beta^*, \delta^* &= \arg\min_{\alpha, \beta, \delta} \sum_{(x_j, \tilde{y}_j) \in \mathcal{D}_v} \mathcal{L}_{W_{\theta^*}} \end{aligned} \tag{4}$$

Crucially, the validation set $\mathcal{D}_v$ does not need to be clean. Accuracy on a held-out noisy set is an unbiased estimator of accuracy under the noisy data distribution [1], so LiLAW can steer training using the same validation partition used for early stopping. This property is essential for our setting, where a clean labelled subset of the EHR data is not available.

*Implementation.* LiLAW wraps the existing training loop without modifying the model architecture, adding only one forward and one backward pass on a validation mini-batch per training batch. Meta-parameters are initialised at $\alpha = 10.0$, $\beta = 2.0$, $\delta = 6.0$ and updated by SGD with learning rate 0.005 and weight decay $10^{-4}$. The first epoch uses standard unweighted BCE loss as a warmup before LiLAW activates, allowing the model to develop an initial difficulty signal before meta-learning begins.

### Combined approach

In the combined condition, LiLAW training is applied to the GraphRAG-imputed dataset. The two interventions target complementary failure modes. GraphRAG corrects the deterministic subset of false negatives, where a documented diagnosis clearly implies a test was needed. LiLAW handles the residual probabilistic uncertainty, down-weighting samples whose loss magnitude during training suggests misclassification. Imputation reduces the noise rate; LiLAW adapts to whatever noise remains.

## Evaluation

### Data partitioning

The cohort is partitioned temporally to simulate prospective deployment: encounters are sorted by arrival date and split sequentially, so the model is always evaluated on data from later in time than it was trained on. Random splitting would allow future information to leak into training through seasonal and longitudinal patterns in the data. Following the split protocol established in the predecessor study [2], 55% of encounters form the training set, 15% the validation set, and 30% the held-out test set, spanning the full cohort window of June 2018 to January 2026.

### Model architecture

All four experimental conditions train and evaluate the same *MultiModalFusionTransformer* architecture inherited from the current deployed MLMD system. Tabular features (vitals, demographics, CTAS score, postal distance, cyclical datetime encodings) are projected from a 27-dimensional input to a 256-dimensional embedding via two linear layers. Three free-text fields (chief complaint and triage note fields) are each encoded independently by a BERT-based biomedical language model served via a local MLflow endpoint, producing three 256-dimensional text embeddings. The four representations (one tabular, three text) form a sequence of tokens processed through four transformer encoder layers with four attention heads and a feedforward dimension of 1024. An attention pooling layer aggregates the sequence into a single vector, which passes through a layer normalisation and a linear classification head for binary prediction. The model has approximately 3.3M trainable parameters.

Text fields are encoded with ClinicalBERT [25], a BERT model pretrained on clinical notes from MIMIC-III discharge summaries.

This architecture replaces the 5-layer fully connected network from the original MLMD validation [2]. The fusion design allows the model to use free-text triage notes directly without the QuickUMLS concept-extraction step required by the predecessor.

### Training configuration

All conditions use identical training hyperparameters to isolate the effect of label strategy. Models are optimised with RAdam (learning rate $10^{-3}$, weight decay $10^{-4}$) with a ReduceLROnPlateau scheduler (minimum learning rate $10^{-4}$, monitored on validation loss). Training runs for up to 15 epochs with early stopping on validation loss. Batch size is 64.

Class imbalance is handled via a per-pathway positive weight applied to the binary cross-entropy loss. The weight is computed as $w^+ = \min\left(\frac{N_{\text{neg}}}{N_{\text{pos}}},\ 10.0\right)$, where the cap of 10.0 prevents extreme weighting for the most imbalanced pathways. Across the four test types, positive-class proportions range from approximately 0.4% (Arm X-Rays) to 2.5% (ECGs), producing raw imbalance ratios between roughly 38:1 and 252:1 before capping.

### Experimental conditions

We compare four training conditions:

1. *Production Baseline*: standard training on the original regex-imputed labels, reproducing the currently deployed configuration.
2. *GraphRAG Imputation Only*: standard training on the KG-imputed dataset (11,110 additional positive labels), with no change to the training objective.
3. *LiLAW Only*: LiLAW-weighted training on the original regex labels, with no change to the label set.
4. *Combined*: LiLAW-weighted training on the KG-imputed dataset.

Conditions (1) and (2) isolate the effect of label correction; conditions (1) and (3) isolate the effect of noise-aware training; condition (4) versus (2) and (3) tests whether the two interventions provide complementary gains.

### Metrics

Recall (true positive rate) is the primary metric. Each missed prediction returns a patient to the serialized queue, so recall directly measures triage efficiency. Precision (positive predictive value) is the safety constraint: unnecessary test authorizations burden ancillary departments and erode clinician trust in the system over time. We report precision-recall (PR) curves across the full range of operating thresholds rather than single-threshold summaries, because clinical deployment requires understanding the full efficiency-safety tradeoff before selecting an operating point.

AUROC captures overall discriminative ability under class imbalance. Expected Calibration Error (ECE) measures whether model confidence scores are reliable enough for clinicians to act on: a well-calibrated system permits threshold selection based on stated probability of need rather than empirical operating-point tuning.

### Imputation evaluation

The imputation pipeline is evaluated separately from the downstream model. Novel positive imputations absent from the production regex baseline are classified as "discoveries": candidate clinical false negatives the regex system missed. The production regex system serves as the high-precision, low-recall reference point; GraphRAG discoveries represent potential recall gains the regex cannot capture.

To validate that discoveries reflect genuine clinical false negatives rather than retrieval errors, a clinical domain expert performed retrospective chart review on a random sample of newly imputed labels; results are reported in Section 6.

## Results

### Knowledge Graph integrity

**[TODO: Alice to draft once KG validation statistics are compiled.]**

### Label imputation performance

Table 1 compares the two imputation strategies against the regex baseline across all four diagnostic pathways.

KG String Matching showed extreme variance across pathways. It performed well on Appendix Ultrasounds ($F_1 = 0.801$), where clinical documentation is standardized enough for lexical rules to succeed, but failed on Arm X-Rays (Recall: 0.031), where arm injury documentation varies too widely for substring matching. On Testicular Ultrasounds, precision collapsed to 0.048 (20,035 discoveries) because anatomical terms in that pathway overlap with unrelated symptom descriptions that a substring match cannot disambiguate.

GraphRAG Semantic Retrieval was substantially more consistent. Operating over semantic similarity rather than lexical overlap, it achieved 0.944 precision on Testicular Ultrasounds and 0.733 recall on Arm X-Rays, recovering the presentations that regex and string matching missed. The exception was Appendix Ultrasounds, where recall dropped to 0.198, likely because the embedding space flattens distinctions that lexical rules preserve in this narrowly documented pathway. Across all four pathways, GraphRAG identified 11,110 total discoveries. Chart review of $n$ = [TODO] sampled discoveries confirmed [TODO]% as genuinely indicated tests, validating that semantic retrieval recovers true clinical false negatives rather than generating systematic hallucinations.

### Downstream MLMD ablation

Table 2 summarizes downstream MLMD performance across the four training conditions on the held-out test set. All values are pending from ablation runs.

Training on GraphRAG-imputed labels exposes the model to 11,110 additional positive examples that the regex baseline misclassified as negatives, broadening recognition of clinical presentations that external-transfer noise had suppressed. Recall increased from [TODO]% (production baseline) to [TODO]% (imputation only).

**[TODO: Tyler to draft once LiLAW ablation runs complete.]**

Both imputation strategies operate within acceptable bounds for offline retrospective use. String matching requires negligible overhead (0.55 ms per encounter, 6.16 ms cold-start initialization). GraphRAG semantic

**Table 1.** Performance of both imputation strategies against the regex baseline. "Discoveries" are novel positive imputations absent from the regex baseline; in this Positive-Unlabeled setting they represent potential clinical false negatives.

| Pathway | Method | Precision | Recall | $F_1$ | Discoveries |
|---|---|---|---|---|---|
| ECG | KG String | 0.786 | 0.871 | 0.827 | 2,900 |
| ECG | GraphRAG | **0.817** | 0.837 | **0.827** | 2,292 |
| Arm X-Ray | KG String | 0.009 | 0.031 | 0.013 | 6,696 |
| Arm X-Ray | GraphRAG | **0.139** | **0.733** | **0.234** | 8,632 |
| Appendix US | KG String | 0.809 | **0.793** | **0.801** | 617 |
| Appendix US | GraphRAG | **0.870** | 0.198 | 0.322 | 97 |
| Testicular US | KG String | 0.048 | 0.352 | 0.085 | 20,035 |
| Testicular US | GraphRAG | **0.944** | **0.521** | **0.672** | 89 |

**Table 2.** Downstream MLMD ablation across four training conditions. Recall is the primary efficiency metric; Precision is the safety constraint. ECE column to be added once calibration measurements are available. **All values pending.**

| Training Condition | Precision | Recall | $F_1$ | AUROC |
|---|---|---|---|---|
| Production Baseline (Regex) | [TODO] | [TODO] | [TODO] | [TODO] |
| GraphRAG Imputation Only | [TODO] | [TODO] | [TODO] | [TODO] |
| LiLAW Only | [TODO] | [TODO] | [TODO] | [TODO] |
| **Combined (GraphRAG + LiLAW)** | **[TODO]** | **[TODO]** | **[TODO]** | **[TODO]** |

retrieval incurs higher cost due to the local biomedical embedding model (175.54 ms per encounter, 17.75 s initialization), but this latency is clinically inconsequential for offline imputation and functionally instantaneous for interactive query during prospective clinical use.

## Discussion and Limitations

The GraphRAG pipeline identified 11,110 test-ordering events that the production regex system misclassified as negatives. The performance gap between retrieval strategies is itself informative. On Testicular Ultrasounds, GraphRAG achieved 0.944 precision where KG String Matching collapsed to 0.048: anatomical terms in that pathway overlap frequently enough with unrelated symptom descriptions to overwhelm a substring match, while biomedical semantic embeddings resolve the ambiguity. The inverse held for Appendix Ultrasounds, where string matching reached 0.793 recall against GraphRAG's 0.198. When documentation for a pathway is sufficiently standardized, lexical matching is reliable and semantic generalization introduces noise rather than resolving it. A pathway-conditional strategy (string matching where documentation is formulaic, GraphRAG where it varies) would be a natural next step.

The downstream ablation will determine whether 11,110 additional positive labels translate to meaningful recall gains at deployment, and whether LiLAW provides additive benefit once the label distribution has been partially corrected. The two interventions target complementary failure modes: data-level correction where clinical evidence is conclusive, algorithmic re-weighting where it is not. Whether LiLAW's marginal gain over imputation alone justifies its training complexity is the central empirical question the pending results will answer.

The MLMD is decision support, not automation. Clinicians retain final authority to override any prediction. This distinction matters for interpreting recall failures: a missed prediction does not mean a missed test, because the physician can still order it during assessment. The cost of a miss is time (the patient returns to the serialized workflow), not patient safety. Precision failures carry a different risk: unnecessary authorizations burden ancillary departments and may erode clinician confidence in the system over time. The evaluation framework reflects this asymmetry, treating recall as the efficiency metric and precision as the safety constraint.

### Limitations

The Knowledge Graph is a point-in-time artifact. RAG over clinical guidelines grounds the initial construction in current evidence, but the vector database is built statically with no automated mechanism for incorporating

guideline updates. Clinical recommendations evolve; a graph that reflects 2025 evidence may diverge from practice within two years without continuous maintenance. The semantic pipeline also relies on a single embedding model (MedEmbed), and sensitivity to alternative biomedical encoders was not systematically characterized.

Label noise cannot be fully corrected by imputation. GraphRAG recovers cases where a post-assessment diagnosis clearly implies a test was needed, but it cannot recover cases where the patient was transferred before any diagnosis was assigned or where the clinical assessment at SickKids differed from the external facility's. LiLAW addresses residual probabilistic noise through the model's loss landscape rather than observed ground truth. Its effectiveness on real clinical asymmetric noise, as opposed to the synthetic benchmarks in prior work [1], is evaluated here for the first time, but external validation at other institutions has not been performed.

Neural network opacity imposes a practical ceiling on clinician adoption. Even with well-calibrated confidence scores, the model cannot explain in clinical terms why it predicts a specific test for a specific patient, limiting the depth of clinical oversight the system can support.

## Future Work

The most immediate priority is making the Knowledge Graph self-maintaining. A continuous integration pipeline that periodically re-scrapes clinical guideline sources, re-indexes the vector database, and flags divergent rules for human review would allow the graph to track evolving evidence without manual reconstruction. Paired with a lightweight approval interface where new rule batches require clinician sign-off before entering the live graph, this would close the human-in-the-loop gap in the current automated pipeline.

With continuous maintenance in place, the natural next step is transitioning from offline retrospective label correction to live prospective decision support at triage. The KG pipeline could infer needed diagnostics from presenting symptoms before physician assessment, directly serving the MLMD's parallelization objective. Computational cost becomes the binding constraint: the GraphRAG pipeline's 175 ms per-encounter latency is acceptable at current SickKids volumes but would require model quantization, vector caching, or distilled biomedical encoders to scale across multi-facility deployments.

At scale, the efficiency gains compound. A 10-percentage-point recall improvement across SickKids' approximately 80,000 annual ED visits would redirect roughly 8,000 test authorizations per year from the serialized to the parallelized pathway. The broader contribution is methodological. Asymmetric label noise from unobserved external clinical events is a systematic problem across fragmented hospital networks, not a SickKids-specific artifact. The Knowledge Graph and meta-learning framework developed here offers a deployable approach for addressing it without cross-institutional EHR integration, a dependency that current interoperability infrastructure cannot satisfy.

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